

Week 4 Narrative Report: Synesthetic Learning Project

Qylle Christian Quino

Department of Data Science - College of Information Technology and Computing
University of Science and Technology of Southern Philippines
Lapasan, Cagayan de Oro City 9000, Philippines

This week focused on comparing several model architectures for the audio emotion classifier and improving the strongest one through hyperparameter tuning. Since the preprocessing pipeline was already in place, the goal was to identify which model handled Mel-spectrogram inputs most effectively and determine the configuration that would carry forward into the cross-modal stages of the project.

I helped evaluate a set of models that included CNN-based, hybrid recurrent, and transformer-style architectures. After running them under the same training conditions, the simple CNN performed best, reaching the highest validation accuracy while remaining the most efficient to train. More complex models like ResNet and the hybrid CNN-LSTM variants took much longer and showed no meaningful improvements, confirming that convolutional features were the best fit for our dataset.

Once the CNN was selected, I used Optuna to run multiple tuning trials. The search tested a range of learning rates, batch sizes, and regularization values. The best trial settled on a moderately small learning rate and a batch size of 16, resulting in a tuned model that reached a validation accuracy a little above 0.82. Class performance was balanced overall, with the model doing particularly well on Sad samples.

To check consistency, I trained the optimized CNN for a full set of epochs and reviewed its learning curves, confusion matrix, and classification metrics. The results confirmed that the model was stable and ready for the next part of the system.

By the end of the week, I also prepared the initial structure for the conditional GAN that will later use the classifier's embeddings. Early training steps showed good loss behavior, indicating that the adversarial setup is functioning properly.